

FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY (FISAT)



Hormis Nagar, Mookkannoor PO, Angamaly, Kochi

Accredited by NAAC with 'A+' Grade

DEPARTMENT OF COMPUTER APPLICATIONS

SEMINAR ABSTRACT

Name of the Student	Bhagya K Ajayan
Batch, Roll Number	A,40
Contact Number	8281255654
Email id	bhagyaajayan@gmail.com
Seminar Topic	Fast Convergence Algorithms For Active Noise Controlling Vechicles

Description of the topic:

Fast Convergence Algorithms for Active Noise Control in Vehicles are advanced techniques designed to enhance the performance of ANC systems, ensuring rapid and efficient noise reduction inside vehicle cabins. ANC systems actively cancel unwanted sounds, such as engine noise, road vibrations, and wind turbulence, by detecting them with microphones and generating counteractive sound waves through speakers. Fast convergence algorithms, like Filtered-X Least Mean Squares (FxLMS), Recursive Least Squares (RLS), and machine learning-based methods, are essential to adapt quickly to dynamic noise conditions typical in vehicles. These algorithms are optimized for real-time processing, ensuring minimal delay in noise cancellation while addressing computational constraints of embedded systems. They are particularly impactful in electric and luxury vehicles, where quieter cabins improve passenger comfort. Future developments focus on incorporating AI for predictive noise cancellation, personalized solutions for different cabin zones, and energy-efficient algorithms tailored to battery-powered vehicles, paving the way for smarter, quieter, and more comfortable automotive experiences.

